MATH 114 HW 4

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Question 1

a. Define $\bar{X} = \frac{1}{n}\sum_{i=1}^{n} X_i$ and $\mathbb{E}(X) = \pi$ since $X_0 \sim \pi, X_i \sim \pi\ \forall i$. We have

$$\begin{aligned}
\mathbb{E}(\bar{X}) &= \mathbb{E}\left(\frac{1}{n}\sum_{i=1}^{n} X_i\right) \\
&= \frac{1}{n}\sum_{i=1}^{n} \mathbb{E}(X_i) \\
&= \frac{1}{n} \cdot n\mathbb{E}(X) \\
&= \mu
\end{aligned}$$

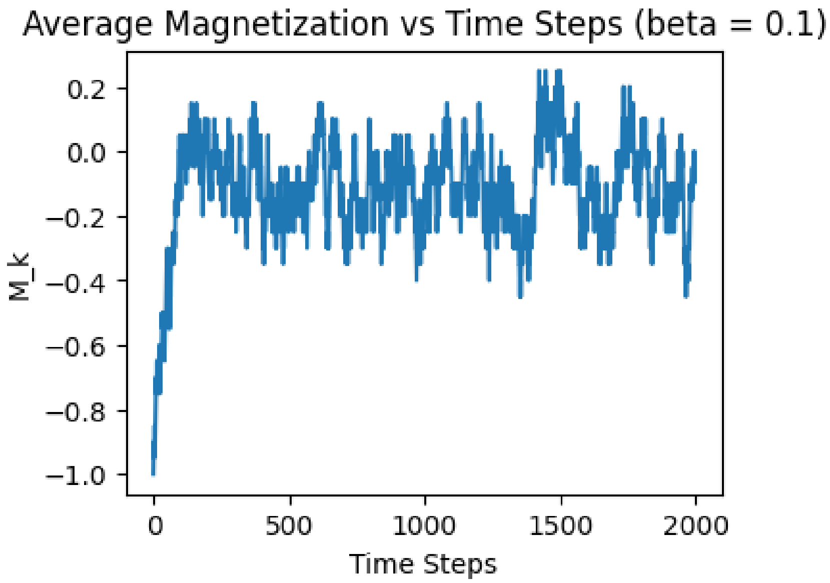
b. We have $\mathbb{P}(X_{t+s} = j, X_t = i) = \mathbb{P}(X_{t+s=j|X_t=i})\mathbb{P}(X_t = i)$

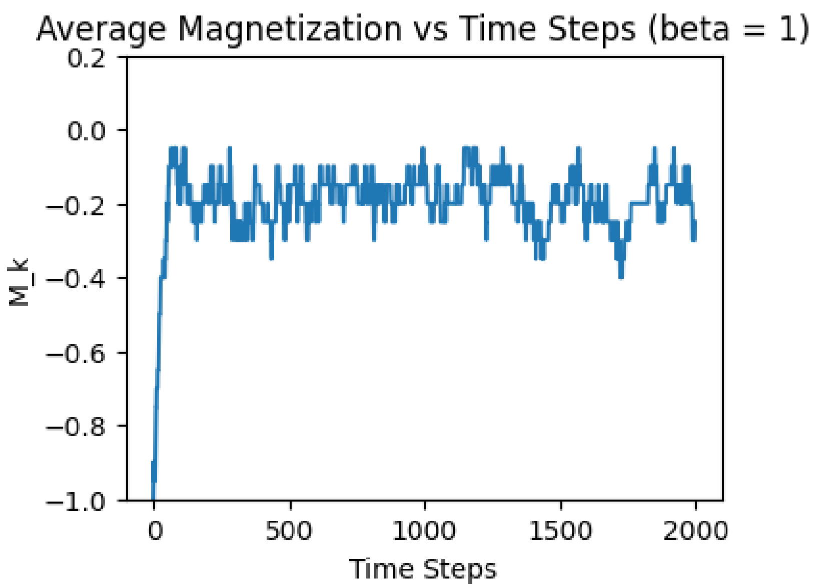
$$\begin{aligned}
\mathbb{E}(X_t X_{t+s}) &= \sum_{i,j=1}^{N} (ij)\mathbb{P}(X_{t+s} = j, X_t = i) \\
&= \sum_{i,j=1}^{N} (ij)\mathbb{P}(X_{t+s} = j | X_t = i)\mathbb{P}(X_t = i) \\
&= \sum_{i,j=1}^{N} (ij)(P_{ij})^s \cdot \pi(i) \\
&= \sum_{i,j=1}^{N} (ij)\mathbb{P}(X_{0+s} = j | X_0 = i)\mathbb{P}(X_0 = i) \\
&= \sum_{i,j=1}^{N} (ij)\mathbb{P}(X_s = j, X_0 = i) \\
&= \mathbb{E}(X_0 X_s)
\end{aligned}$$

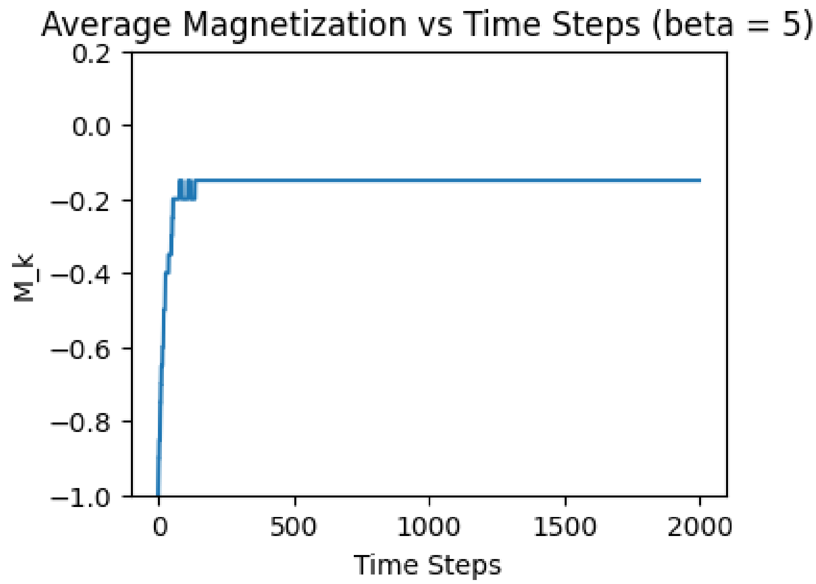
c. $\mathrm{Var}\bar{X} \neq \frac{1}{n}\mathrm{Var}X$ because X_i depends on X_{i-1}, therefore they are not independent, and we can not compute the sum $\sum_{i=1}^{n} \mathrm{Var}X_i$.

Question 2

a. i.





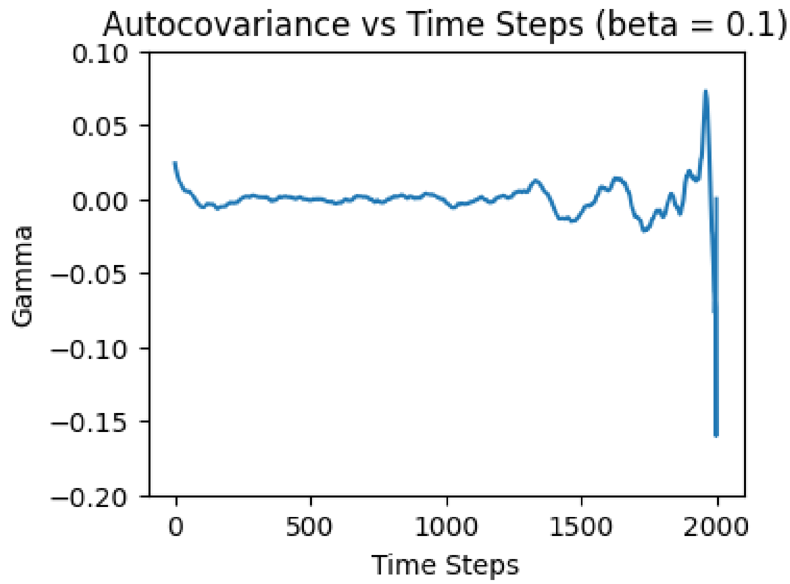


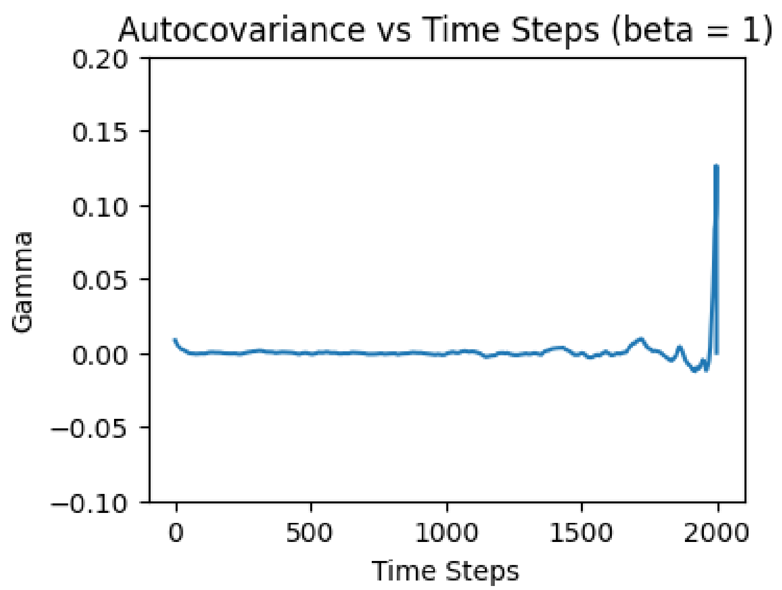
Note that

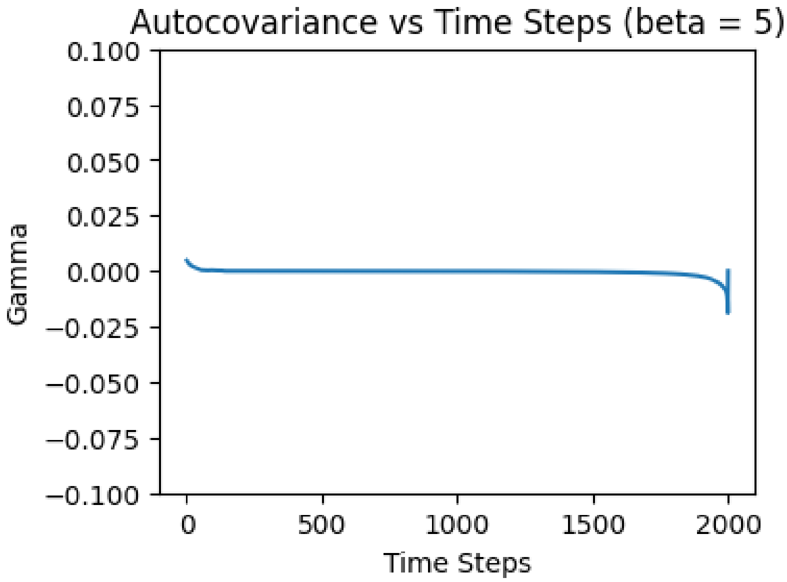
$$\alpha = \frac{f(W)}{f(X_i)} = \frac{\exp(-\beta H(W))}{\exp(-\beta H(X_i))} = \exp(-\beta(H(W) - H(X_i)))$$

Therefore the bigger the β, the lower α, and the more likely we accept our proposal state and the less that our average magnetization will fluctuate. This matches what we see within the graphs, where the higher the β's, the less the graph fluctuates around -0.15.

ii.


Autocovariance vs Time Steps (beta = 0.1)
0.10
0.05
0.00
−0.05
−0.10
−0.15
−0.20
Gamma
0
500
1000
1500
2000
Time Steps


Autocovariance vs Time Steps (beta = 1)
0.20
0.15
0.10
0.05
0.00
−0.05
−0.10
Gamma
0
500
1000
1500
2000
Time Steps



Autocovariance seems to decrease slightly as we added more time steps to the simulation, however there is a weird spike at the end of the simulation around 2000 time steps.

b. i.

Question 3